



Computational Fluid Dynamics- Fundamentals

Exercise 1: Nozzle Analysis

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1 Introduction

1.1 Geometrical Domain

Geometrical domain data are here presented:

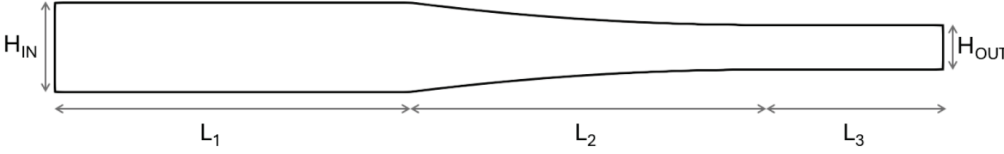


Figure 1.1: Nozzle geometry

Data	Value [m]
L_1	1
L_2	1
L_3	0.5
H_{in}	0.25
H_{out}	0.125

Table 1.1: Geometric dimensions

The two rectangular blocks are connected by two nozzle-shaped arcs, tangent to the top and bottom lines of the outlet block and passing through the points of the inlet block. The points of each arc are determined by solving, in each case, the equation of a circle that passes through the two points previously described, knowing that the x coordinate of the center corresponds to 2 due to tangency, the radius of the circumference remains the only unknown. So, having the two circumference equations, two middle points for the upper and lower arc are determined: $(1.5, 0.0469, 0)$ and $(1.5, 0.2031, 0)$, in order to use them in arcs definition of *system/blockMeshDict*.

1.2 Physical flow

The fluid under study is an incompressible air in ambient conditions ($\nu = 15 \times 10^{-6} \text{ m}^2/\text{s}$, $\rho = 1.2 \text{ kg/m}^3$). The jump of pressure from the inlet to the outlet is $P_{T,in} - P_{out} = 0.0096 \text{ Pa}$, which is quite low. So, we expect the flow to enter the nozzle, creating a boundary layer, and to evolve in a stable flow, which will encounter the smooth restriction of the channel. At the beginning of the outer block, the flow, exiting from the convergent section, will start again to reach a totally developed form. In order to start entering the problem and to have an idea of the results that the simulations will show us, here are presented calculations considering an isentropic converging nozzle, in which the total pressure is conserved:

$$P_{t,in} = P_{t,out} = 101325.0096 \text{ Pa}$$

From the total pressure equation at the outlet:

$$P_{t,out} = P_{out} + \frac{1}{2} \rho v_{out, is}^2$$

Assuming:

$$P_{out} = P_{amb} = 101325 \text{ Pa}$$

We can solve for the isentropic outlet velocity:

$$v_{out, is} = \sqrt{\frac{2(P_{t,out} - P_{out})}{\rho}} = 0.12469 \text{ m s}^{-1}$$

Finally, the isentropic inlet velocity, by imposing mass conservation, will be:

$$v_{in, is} = \frac{1}{2} v_{out, is} = 0.0632 \text{ m s}^{-1}$$

2 Laminar Flow

2.1 Simulation Setting

In this chapter are reported and commented the chosen settings valid for all laminar simulations. The simulations were initially performed using low-accuracy methods for validation purposes, and subsequently refined with high-accuracy approaches.

Boundary conditions for both velocity and pressure fields, defined in the **0/** folder, are summarized below:

- Boundary conditions - Velocity field:
 - Inlet: type *pressureInletVelocity*, with a uniform value of (0.0632, 0, 0) m/s. The isentropic velocity has been assigned as initial value.
 - Outlet: type *zeroGradient*.
 - TopWall and BottomWall: type *noSlip*.
 - Front and Back: type *empty*, consistent with a 2D simulation.
- Boundary conditions - Pressure field:
 - Inlet: type *totalPressure*, with a kinematic pressure value of $P_0 = 0.008 \text{ m}^2/\text{s}^2$.
 - Outlet: type *fixedValue*, with a uniform value of $0 \text{ m}^2/\text{s}^2$ (which is the actual atmospheric kinematic pressure).
 - TopWall and BottomWall: type *zeroGradient*.
 - Front and Back: type *empty*, consistent with a 2D simulation.

In order to solve PDEs, the mathematical operators must be redefined to work in a discrete way by domain discretization. The numerical schemes chosen for the different terms of the NVS equations are specified inside **fvSchemes** file. The system of equations that describes the problem is solved by the **SIMPLE** algorithm, a pressure-based method specific for steady state simulations. The residual control that defines the convergence of the simulation is set on 10^{-6} for both U and p. Solvers adopted for the resolution of the linear algebraic systems specific for each variable are specified inside the **fvSolution** .

system/fvSchemes	Numerical scheme
ddtSchemes	steadyState
gradSchemes:grad(p)	Gauss linear
div(phi,U)	bounded Gauss linearUpwind grad(U)
laplacianSchemes	Gauss linear corrected

Table 2.1: fvSchemes definition

system/fvSolution	solver	smoother	tollerance
p	GAMG	GaussSeidel	10^{-8}
U	smoothSolver	GaussSeidel	10^{-8}

Table 2.2: fvSolution definition

The information on the *incompressibility* of the flow is inserted in **system/controlDict**. In **constant** folder, **MomentumTransport** and **physicalProperties** files allow respectively to define that the simulated flow is laminar and its values of ν and ρ . In all the cases presented, convergence will be reached within the **endTime** of 3000 outer iterations. Standard relaxation factors have been used.

2.2 First Mesh

2.2.1 Mesh Generation

The domain has been divided in 3 blocks: inlet, convergent and outlet sections. The interfaces must be conformal: for this reason all the blocks have the same amount of cells along y-direction, with the same grading. The patches of the domain are: *BottomWall*, *TopWall*, *Inlet*, *Outlet*, *Front*, *Back*.

Being a laminar flow there are no constraints on the dimensions of the cells close to the walls. The coarsest mesh is characterized by 145 cells along x, 15 along y and 1 along z because the flow is 2D. As it is evident from the reported image, a symmetric multi-gradient has been employed in vertical direction that refines the mesh close to the walls in order to better simulate the behavior in these zones. The convergent block presents a bigger amount of cells along the horizontal direction because it is where the flow changes due to the walls influence. The mesh presents a max non-orthogonality value of 6.85, an average one of 2.14, so a **NonOrthogonalCorrectors** has been added to the SIMPLE. The max skewness is 0,12, while the max aspect ratio is 8,87.

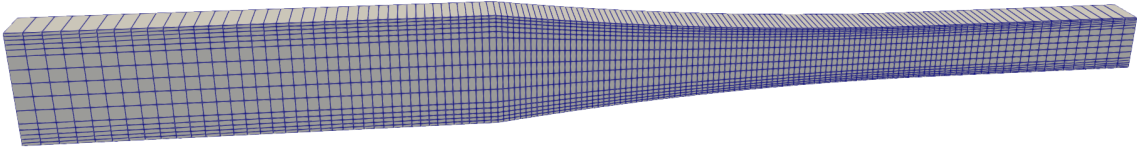


Figure 2.1: Coarse mesh, laminar case

2.2.2 Results

The process has reached convergence in 287 iterations. Plots of velocity and pressure are here reported:

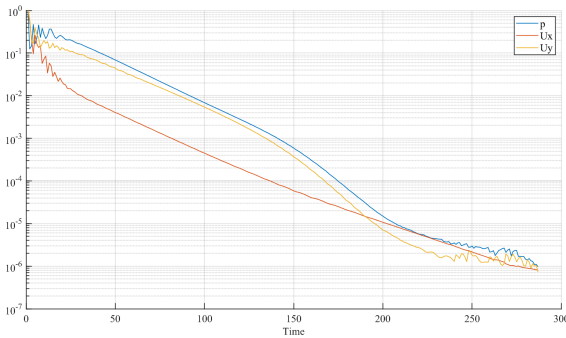


Figure 2.2: Residual trends

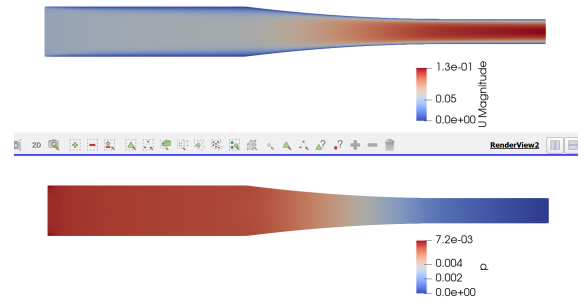


Figure 2.3: U and P

The velocity profiles in three different sections are reported:

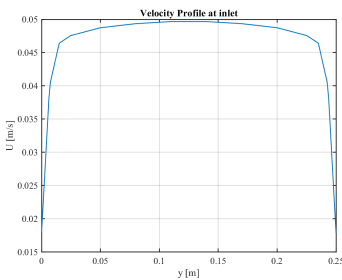


Figure 2.4: Inlet

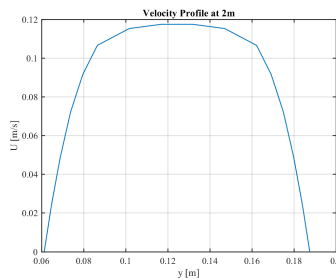


Figure 2.5: x=2m

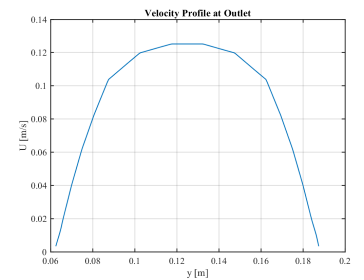


Figure 2.6: Outlet

By integration over the section the mean velocity at the outlet has been computed, $v_{out,m} = 0,0942m/s$. The flow rate per unit width and the velocity reduction coefficient are:

$$Q = H_{out} * v_{out,m} = 0,011775m^2/s; \quad \varphi = \frac{v_{out,m}}{v_{out,is}} = 0,7555$$

The longitudinal distribution of cross-sectional-averaged total, static pressure and velocity along the nozzle are:

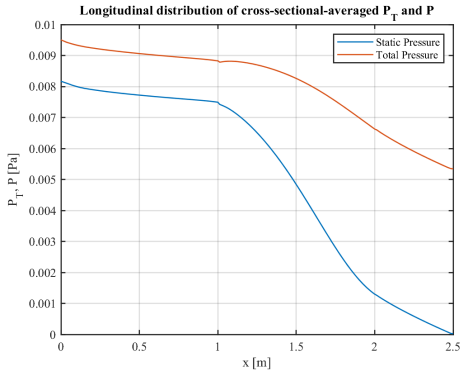


Figure 2.7: $P_{T,m}$ and $P_{s,m}$ distribution

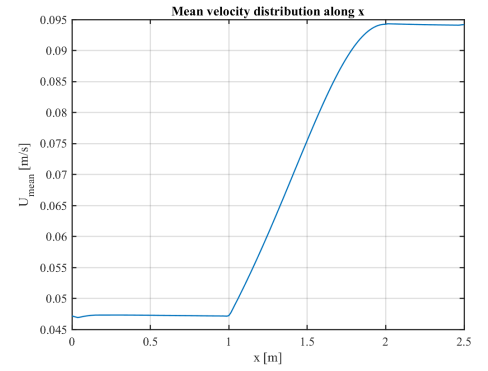


Figure 2.8: Velocity distribution

The distribution of wall shear stress along the Top and Bottom Wall patches is:

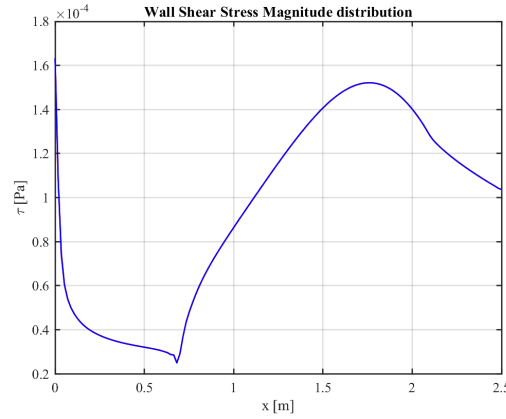


Figure 2.9: τ_w distribution

The stresses, as expected, increase in the convergent part due to the higher velocity gradient generated by the wall effect.

2.3 Second Mesh

2.3.1 Mesh Generation

The second mesh has the same basis of the coarse one; it has been refined by simply increasing the amount of cells both along x and y and by emphasize the gradient effect with the amplified variation of the aspect ratio.

The mesh has: 950 x 80 x 1 cells. It presents a max non-orthogonality value of 7.11, an average one of 2.08, so a **NonOrthogonalCorrectors** has been added as explained before. The max skewness is 0.13, while the max aspect ratio is 10.64.

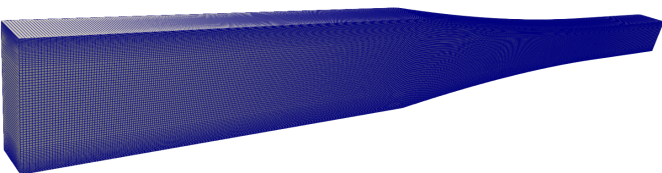


Figure 2.10: Refined mesh, laminar case

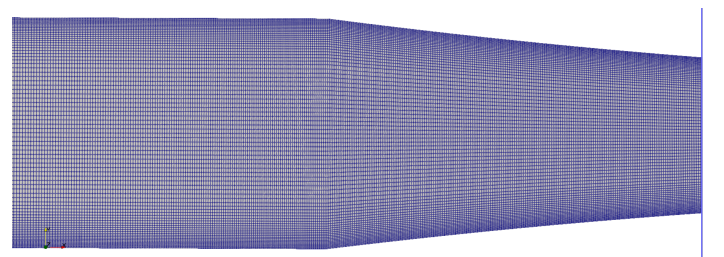


Figure 2.11: Refined mesh, laminar case

2.3.2 Results

The process has reached convergence in 1694 iterations. Plots of velocity and pressure are here reported:

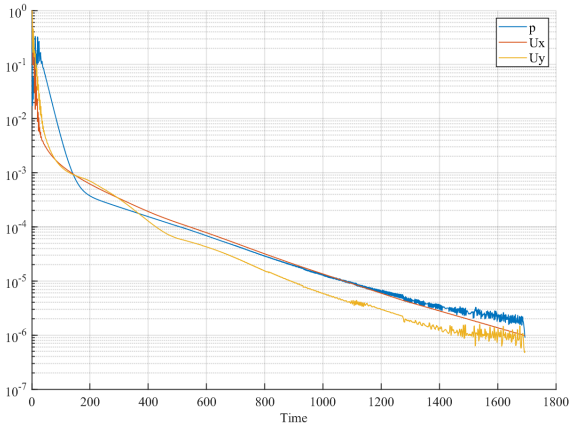


Figure 2.12: Residual trends

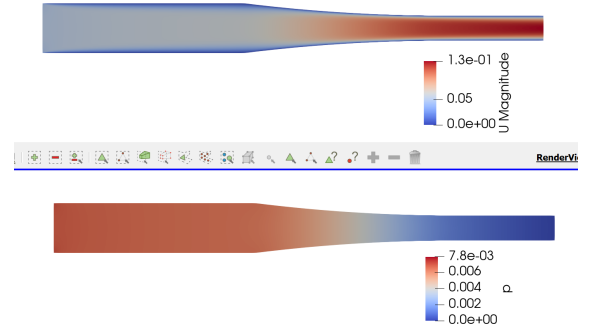


Figure 2.13: U and P

The velocity profiles in three different sections are reported:

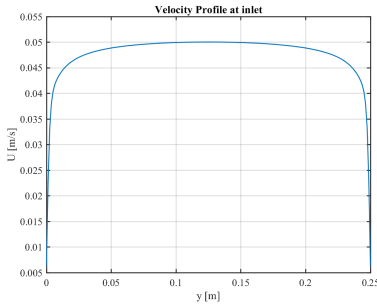


Figure 2.14: Inlet

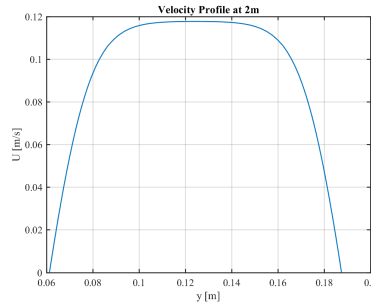


Figure 2.15: x=2m

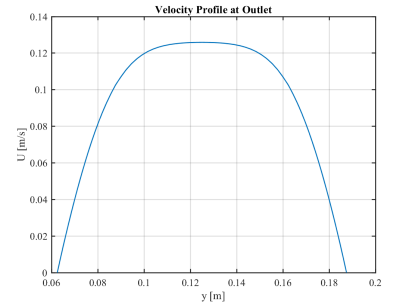


Figure 2.16: Outlet

By integration over the section the mean velocity at the outlet has been computed, $v_{out,m} = 0,0951 \text{ m/s}$. The flow rate and the velocity reduction coefficient are:

$$Q = H_{out} * v_{out,m} = 0,0118875 \text{ m}^2/\text{s}; \quad \varphi = \frac{v_{out,m}}{v_{out,is}} = 0,7626$$

The longitudinal distribution of cross-sectional-averaged total, static pressure along the nozzle and velocity are:

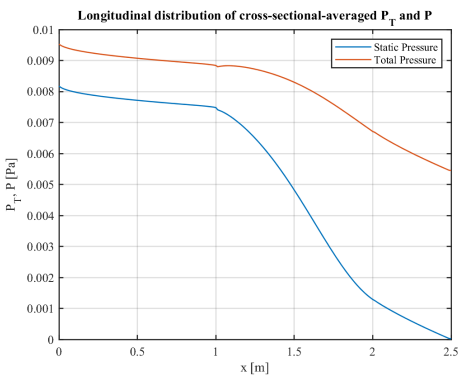


Figure 2.17: $P_{T,m}$ and $P_{s,m}$ distribution

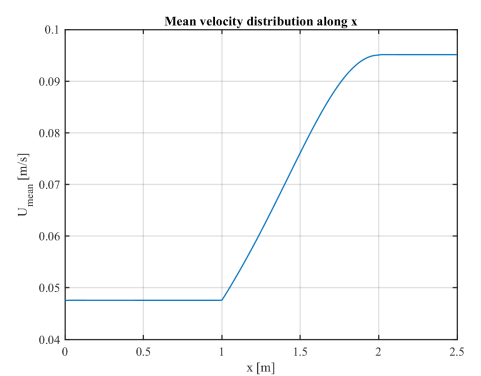


Figure 2.18: Velocity distribution

The distribution of wall shear stress along the Top and Bottom Wall patches is:

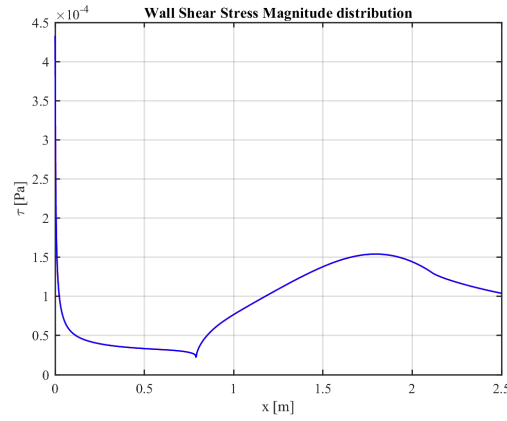


Figure 2.19: τ_w distribution

The stresses increase in the convergent part due to the higher velocity gradient generated by the wall effect.

2.4 Third Mesh

2.4.1 Mesh Generation

The most refined mesh has been obtained by following the same procedure of the others. Even in this case, the number of cells has been incremented, and the grading along y has been raised. The mesh has: 1200x120x1 cells. It presents a max non-orthogonality value of 7.11, an average one of 1.92, so a **NonOrthogonalCorrectors** has been added as explained before. The max skewness is 0.12, while the max aspect ratio is 9.56.

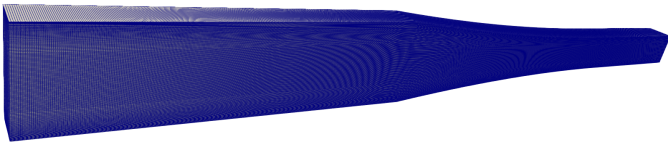


Figure 2.20: More refined mesh, laminar case

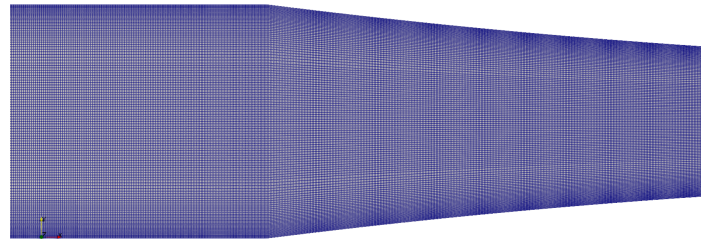


Figure 2.21: More refined mesh, laminar case

2.4.2 Results

The process has reached convergence in 2185 iterations, plots of velocity and pressure are here reported:

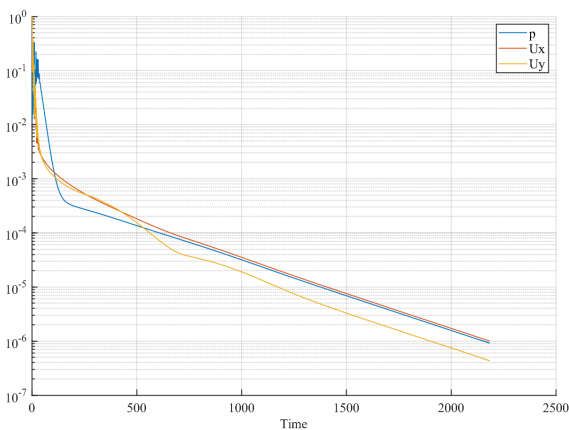


Figure 2.22: Residual trends

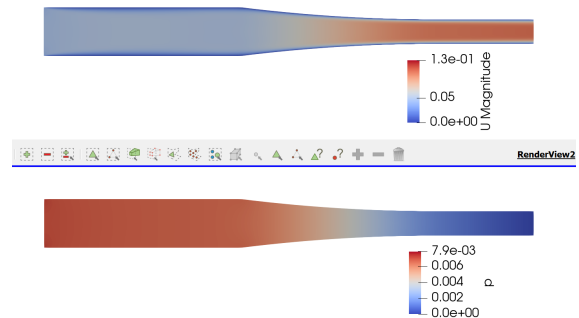


Figure 2.23: U and P

The velocity profiles in three different sections are reported:

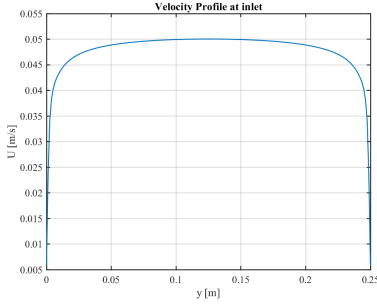


Figure 2.24: Inlet

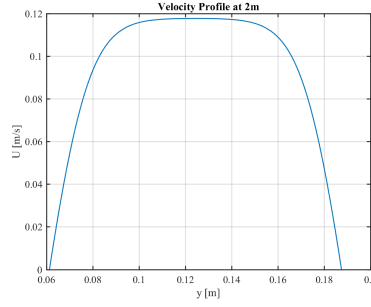


Figure 2.25: x=2m

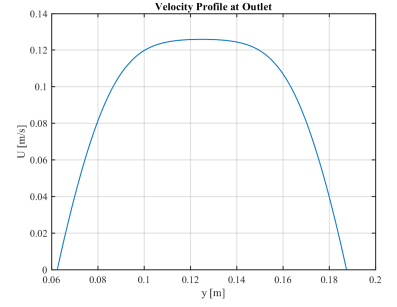


Figure 2.26: Outlet

By integration over the section the mean velocity at the outlet has been computed, $v_{out,m} = 0,0951 m/s$. The flow rate and the velocity reduction coefficient are:

$$Q = H_{out} * v_{out,m} = 0,0118875 m^2/s; \quad \varphi = \frac{v_{out,m}}{v_{out,is}} = 0,7626$$

The longitudinal distribution of cross-sectional-averaged total,static pressure along the nozzle and velocity are:

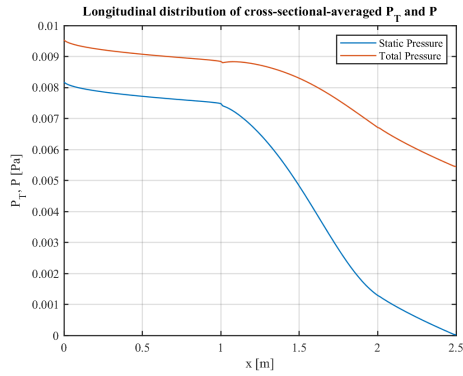


Figure 2.27: $P_{T,m}$ and $P_{s,m}$ distribution

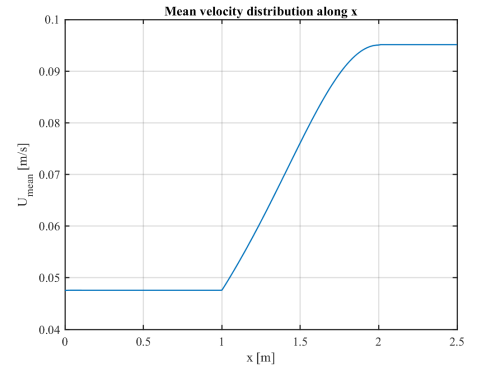


Figure 2.28: Velocity distribution

The distribution of wall shear stress along the Top and Bottom Wall patches is:

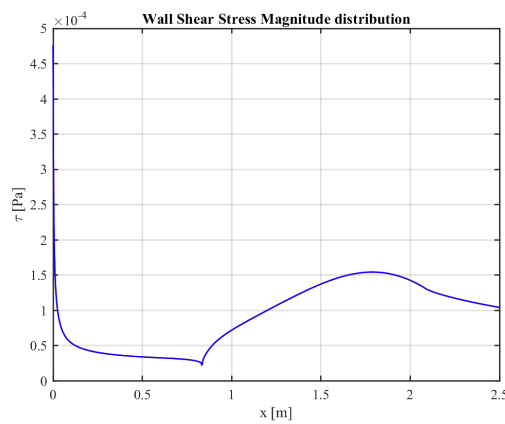


Figure 2.29: τ_w distribution

The stresses increase in the convergent part due to the higher velocity gradient generated by the wall effect.

2.5 Grid Independence

The evaluation of the grid independence of the three simulations has been made by the comparison of the different obtained results.

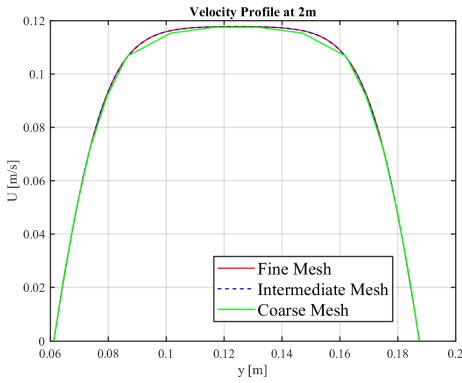


Figure 2.30: velocity at $x=2m$

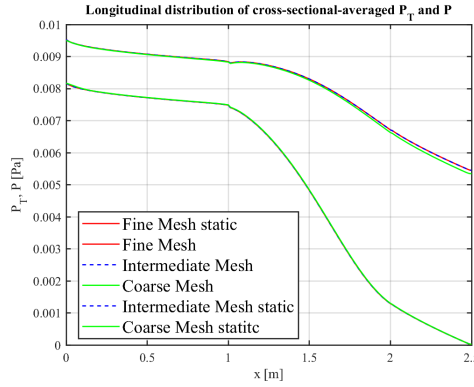


Figure 2.31: Pressure distribution

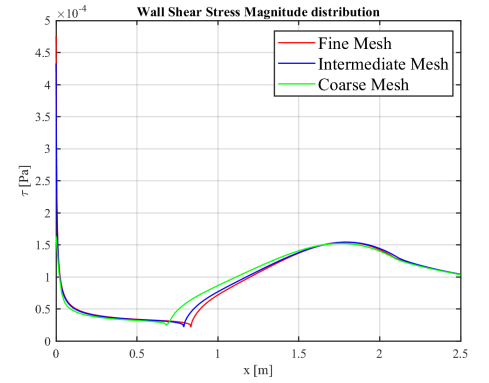


Figure 2.32: Stress distribution

The reported figures clearly show that the Fine and the Intermediate meshes have reached the grid independence; in fact their results are equal and the relative plots completely overlap. The coarse mesh results are slightly different and this can be seen mainly from the velocity profile, this is due to the less accurate refinement close to the walls of the nozzle. In general all the three simulations lead to very similar results even if their number of cells differs a lot, one of the reasons could be the simplicity of the analyzed flow.

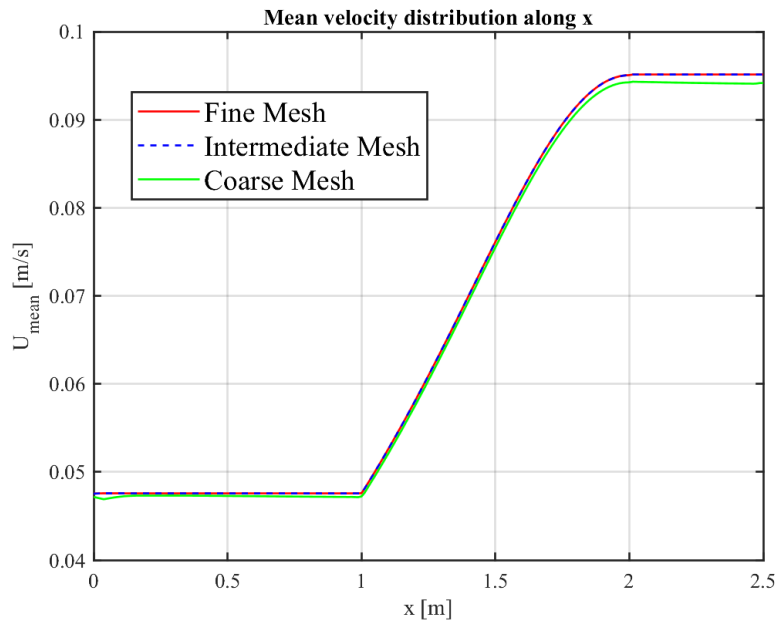


Figure 2.33: Mean U distribution along the nozzle

3 Turbulent Flow: Wall Function

3.1 Simulation Setting

The flow under study is now turbulent and still incompressible; this means that a proper turbulent model must be defined. The request is to use a 2-equations model combined with the use of the wall functions, for these reasons it has been decided to employ the **k-ε** that works well far from the walls. Differently from the previous case now we have two more equations and two more unknown quantities for which specific BCs, numerical schemes and solvers must be defined. The pressure jump from the inlet to the outlet is $P_{T_{in}} - P_{out} = 129,34 \text{ Pa}$ and following the already described procedure the isentropic inlet velocity is found: $v_{in,is} = 7,34 \text{ m/s}$. By knowing the turbulence intensity $I_{in} = 5\%$ and the mixing length $l_T = 0,025 \text{ m}$, it is possible to compute the inlet values for k and ϵ (these values are assigned to the internal field as initial guess):

$$\kappa_{in} = \frac{3}{2} U^2 I^2 = 0,202 \text{ m}^2/\text{s}^2; \quad \epsilon_{in} = 0.09^{3/4} \frac{\kappa^{3/2}}{l_T} = 0,5967 \text{ m}^2/\text{s}^3; \quad \nu_{T,in} = 0.09 \frac{\kappa^2}{\epsilon} = 0,00615 \text{ m}^2/\text{s}^2$$

Boundary conditions for both velocity and pressure fields are of the same type of the laminar case but with the right values of $v_{in,is}$ and $P_{T,in}$. BCs for k, ϵ, ν_T defined in the **0/** folder are summarized below:

0: BCs	k	ε	ν_T
Top & Bottom Wall	type: kqRWallFunction value: uniform \$inletk	type: epsilonWallFunction value: uniform \$inletEpsilon	type: nutkWallFunction value: uniform 0.00615
Inlet	type: turbulentIntensityKineticEnergyInlet intensity: 0.05 value: uniform \$inletk	type: turbulentMixingLengthDissipationRateInlet mixingLength: 0.025 value: uniform \$inletEpsilon	type: fixedValue value: uniform 0.00615
Outlet	zeroGradient	zeroGradient	zeroGradient
Front / Back	empty	empty	empty

Table 3.1: 0/k, 0/epsilon, 0/nut

Even in this case the solution algorithm for the system resolution is the **SIMPLE**, the difference is that now the residual control regards also the quantities k and ϵ . Convergence is reached when all the residuals fall below 10^{-6} . Having two additional equations means that additional solvers must be define inside the **fvSolution** file, and the discretization of the new terms in **fvSchemes**.

system/fvSchemes	Numerical scheme
ddtSchemes	steadyState
gradSchemes:grad(p)	Gauss linear
div(phi,U)	bounded Gauss linearUpwind grad(U)
div(phi,epsilon)	bounded Gauss upwind
div(phi,k)	bounded Gauss upwind
laplacianSchemes	Gauss linear corrected

Table 3.2: fvSchemes definition

system/fvSolution	solver	smoother	tolerance
p	GAMG	GaussSeidel	10^{-8}
U	smoothSolver	GaussSeidel	10^{-8}
"k—epsilon"	smoothSolver	GaussSeidel	10^{-8}

Table 3.3: fvSolution definition

The information on the turbulence model adopted is present in **constant/MomentumTransport:**
simulationType RAS, model kEpsilon.

3.2 First Mesh

3.2.1 Mesh Generation

The mesh has the same blocks with same patches previously seen. However, being now the flow turbulent, precautions must be taken. With kEpsilon model and wall functions, an high y^+ (> 30) has to be guaranteed, with the centroid of the first cell sufficiently far from the wall. To permit this, 20 cells were used along y, with a quite low expansion ratio in *SimpleGrading* : 2 for 20% of cells both on top and on bottom wall. The coarsest mesh here presented is characterized by 475 cells along x, 20 along y and 1 along z because the flow is 2D. The convergent block presents a bigger amount of cells along the horizontal direction, as seen for the laminar case. The mesh presents a max non-orthogonality value of 6.96, an average one of 2.02, so a **NonOrthogonalCorrectors** has been added to the SIMPLE. The max skewness is 0,18, while the max aspect ratio is 4,77.

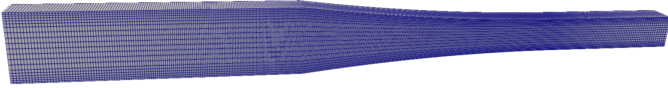


Figure 3.1: Coarse mesh, KEpsilon case

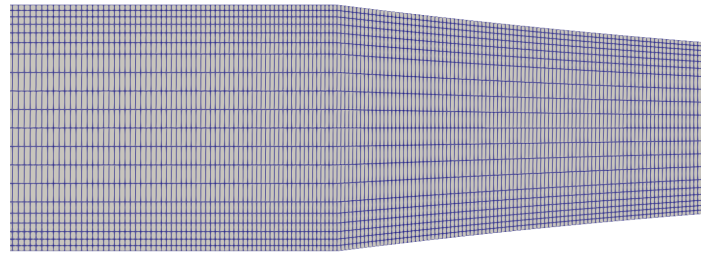


Figure 3.2: Coarse zoomed mesh, KEpsilon case

3.2.2 Results

The process has reached convergence in 1154 iterations. Plots of velocity and pressure are here reported:

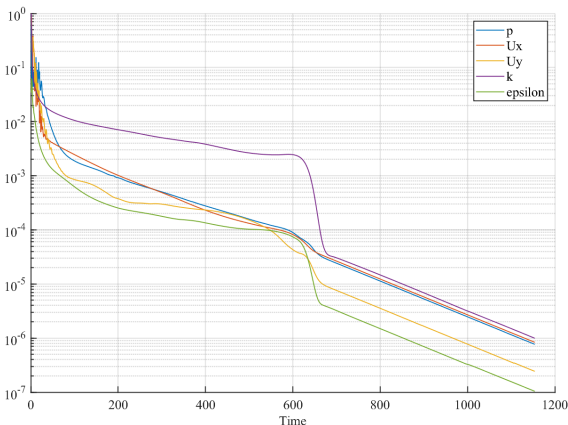


Figure 3.3: Residual trends

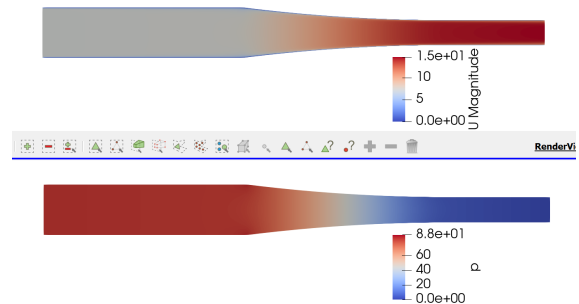
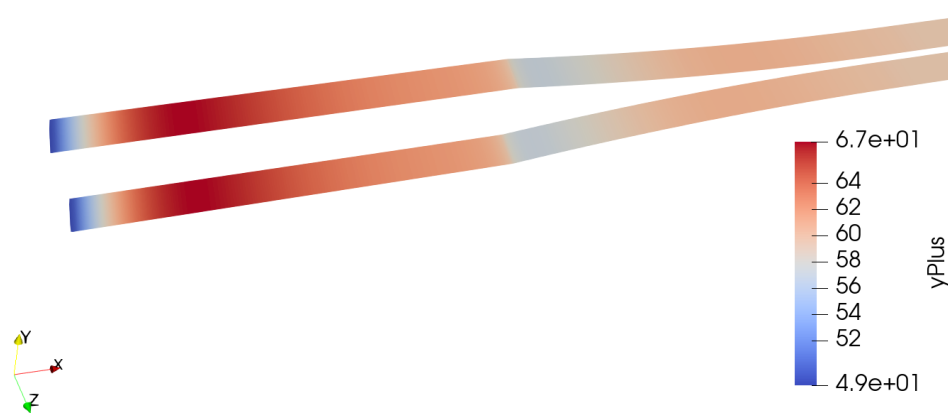


Figure 3.4: U and P

The y^+ value has to be checked. In folder *postProcessing/yPlus/0* at convergence, a minimum value of 48.58 is given, with an average of 60.46 (in both cases beyond the limit value prefixed). A graph representing y^+ located at patches is here reported:

Figure 3.5: y^+ distribution

The velocity profiles in three different sections are reported:

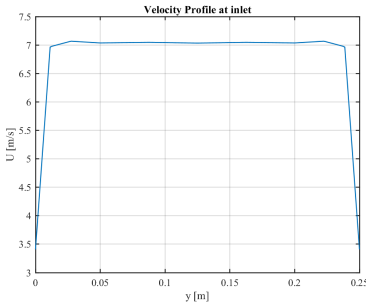


Figure 3.6: Inlet

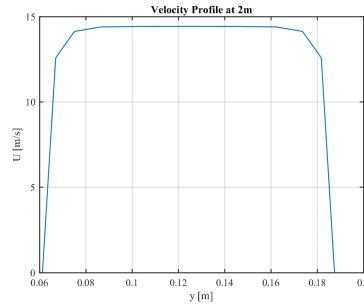
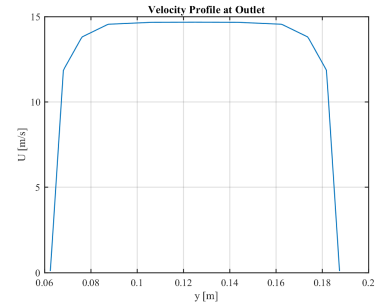
Figure 3.7: $x=2m$ 

Figure 3.8: Outlet

By integration over the section the mean velocity at the outlet has been computed, $v_{out,m} = 13,4685m/s$. The flow rate and the velocity reduction coefficient are:

$$Q = H_{out} * v_{out,m} = 1,6835625m^2/s; \quad \varphi = \frac{v_{out,m}}{v_{out,is}} = 0,9171$$

The longitudinal distribution of cross-sectional-averaged total,static pressure along the nozzle and velocity are:

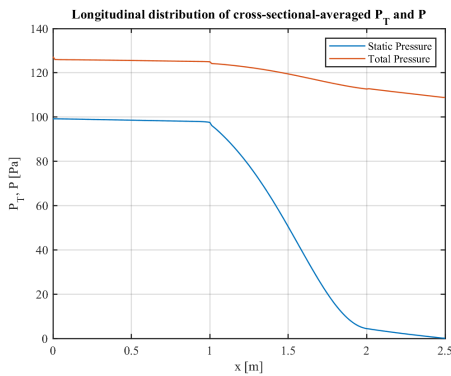
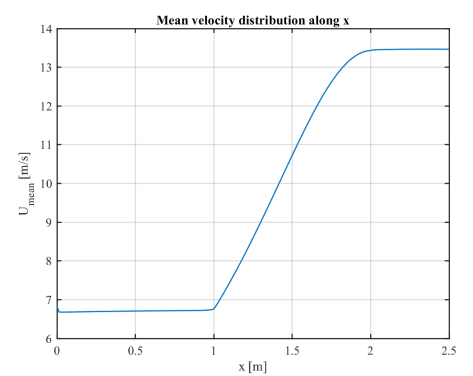
Figure 3.9: $P_{T,m}$ and $P_{s,m}$ distribution

Figure 3.10: Velocity distribution

The distribution of wall shear stress along the Top and Bottom Wall patches is:

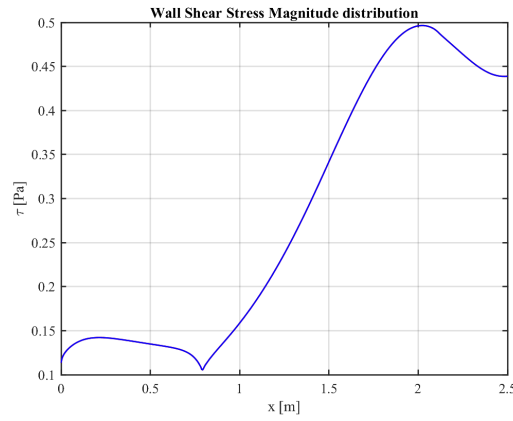


Figure 3.11: τ_w distribution

The stresses increase in the convergent part due to the higher velocity gradient generated by the wall effect.

3.3 Second Mesh

3.3.1 Mesh Generation

The second mesh has the same basis of the coarse one; it has been refined by simply increasing the amount of cells both along x and y . To respect the y^+ limit, the *SimpleGrading* along y has been reduced to 1.5 for 30% of cells both on top and on bottom wall. The mesh has now $800 \times 35 \times 1$ cells. It presents a max non-orthogonality value of 7.03, an average one of 1.99, so a **NonOrthogonalCorrectors** has been added as explained before. The max skewness is 0.18, while the max aspect ratio is 4.72.

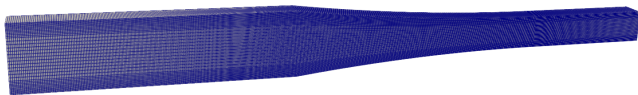


Figure 3.12: Intermediate mesh, KEpsilon case

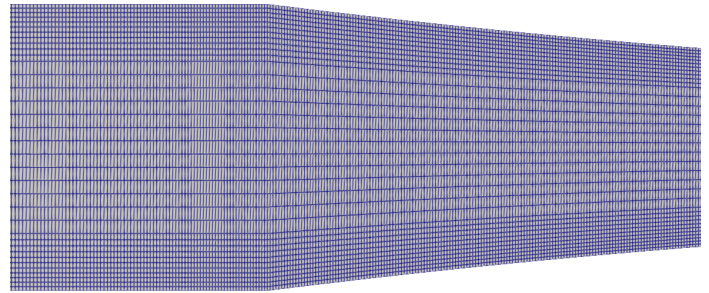


Figure 3.13: Intermediate mesh zoomed, KEpsilon case

3.3.2 Results

The process has reached convergence in 2574 iterations. Plots of velocity and pressure are here reported:

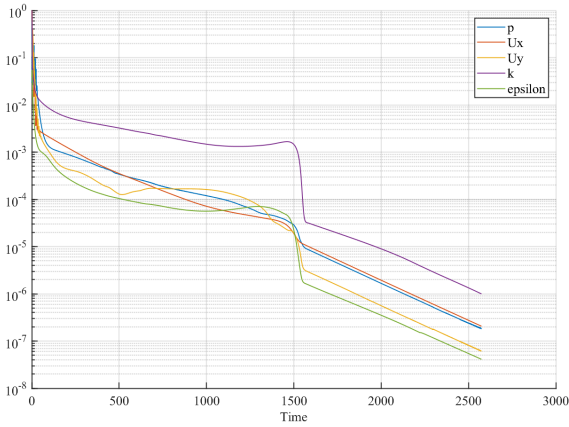


Figure 3.14: Residual trends

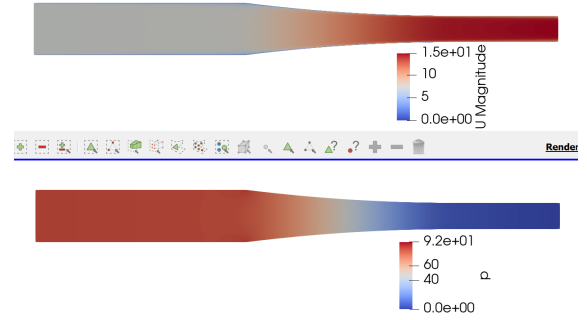


Figure 3.15: U and P

In folder *postProcessing/yPlus/0* at convergence, a minimum value of 28.4 localized at the inlet, and an average of 39.6 (the values between top and bottom wall are very similar, so only one of them is reported). A graph representing y^+ located at that patches is here reported:



Figure 3.16: y^+ distribution

It can be noticed that the minimum y^+ is only located at the inlet of the nozzle. Anyway, that minimum is not relevant and physical: the boundary layer is not fully developed and the turbulent model used in the simulation does not represent its real laminar behavior there. The velocity profiles in three different sections are reported:

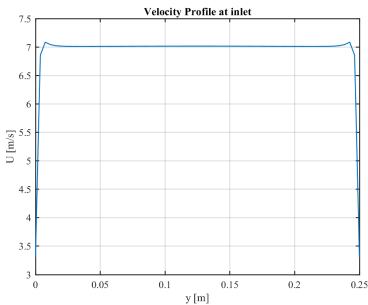


Figure 3.17: Inlet

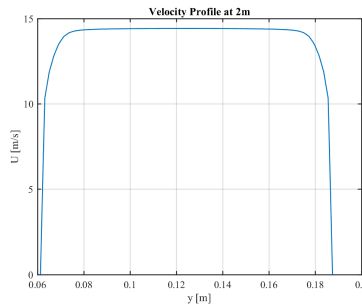


Figure 3.18: x=2m

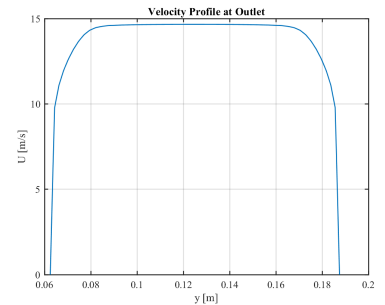


Figure 3.19: Outlet

By integration over the section the mean velocity at the outlet has been computed, $v_{out,m} = 13,8069 \text{ m/s}$. The flow rate and the velocity reduction coefficient are:

$$Q = H_{out} * v_{out,m} = 1,7258625 \text{ m}^3/\text{s}; \quad \varphi = \frac{v_{out,m}}{v_{out,is}} = 0,9404$$

The longitudinal distribution of cross-sectional-averaged total,static pressure along the nozzle and velocity are:

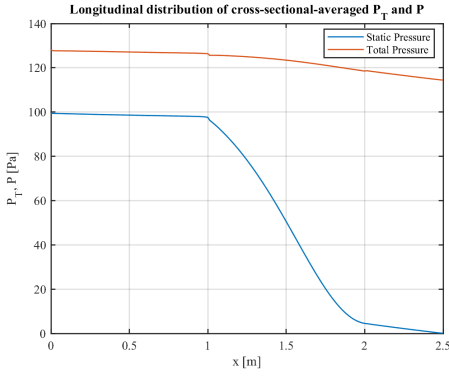


Figure 3.20: $P_{T,m}$ and $P_{s,m}$ distribution

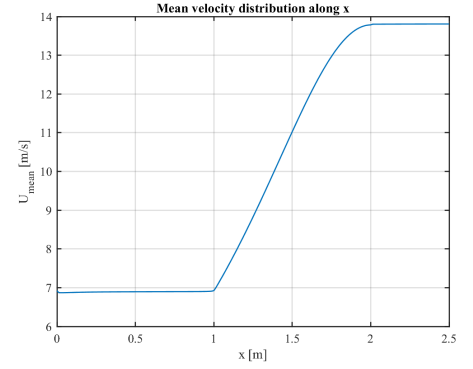


Figure 3.21: Velocity distribution

The distribution of wall shear stress along the Top and Bottom Wall patches is:

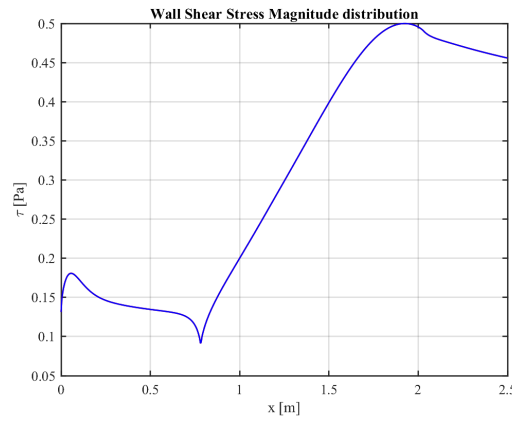


Figure 3.22: τ_w distribution

The stresses increase in the convergent part due to the higher velocity gradient generated by the wall effect.

3.4 Third Mesh

3.4.1 Mesh Generation

The fine mesh follows the topology of the previous ones. It has been refined by simply increasing the amount of cells both along x and y. To respect the y^+ limit, the *SimpleGrading* along y has been reduced to 1.6 for 30% of cells both on top and on bottom wall. The mesh has now 950 x 40 x 1 cells. It presents a max non-orthogonality value of 7.04, an average one of 2.01, so a **NonOrthogonalCorrectors** has been added as explained before. The max skewness is 0.20, while the max aspect ratio is 4.78.

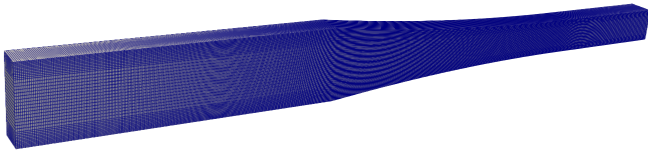


Figure 3.23: Fine mesh, KEpsilon case

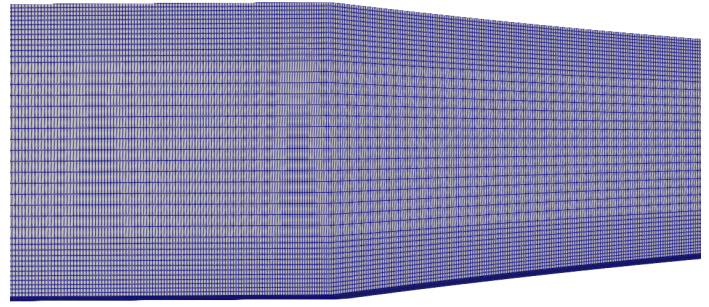


Figure 3.24: Fine mesh zoomed, KEpsilon case

3.4.2 Results

The process has reached convergence in 3251 iterations. Plots of velocity and pressure are here reported:

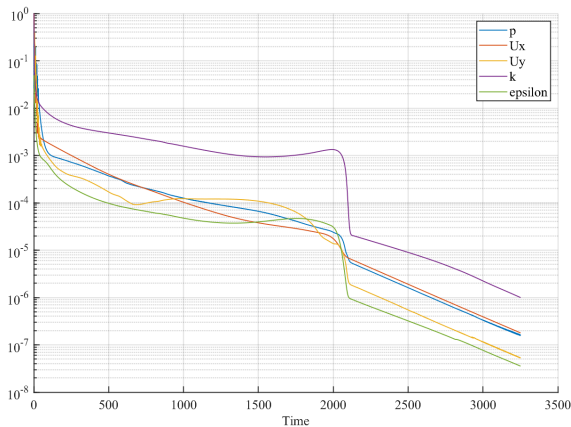


Figure 3.25: Residual trends

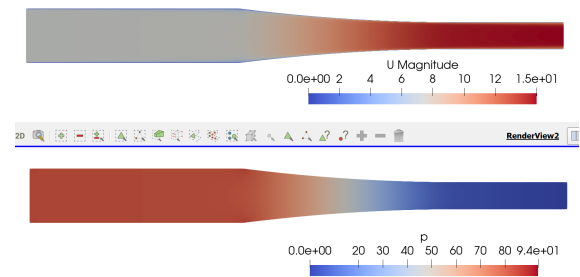
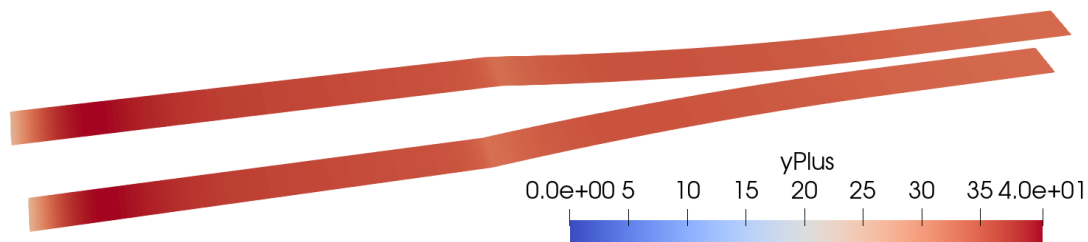


Figure 3.26: U and P

In folder *postProcessing/yPlus/0* at convergence, a minimum value of 24.9 is given, with an average of 35.2 (the values between top and bottom wall are very similar, so only one of them is reported). A graph representing y^+ located at that patches is here reported:

Figure 3.27: y^+ distribution

It can be noticed that the minimum y^+ is only located at the inlet of the nozzle. Anyway, that minimum is not relevant and physical, as explained before.

The velocity profiles in three different sections are reported:

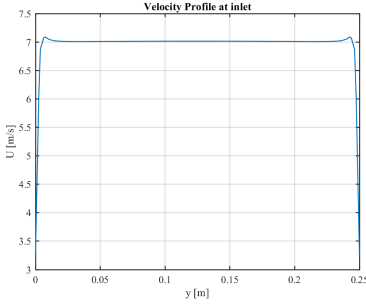


Figure 3.28: Inlet

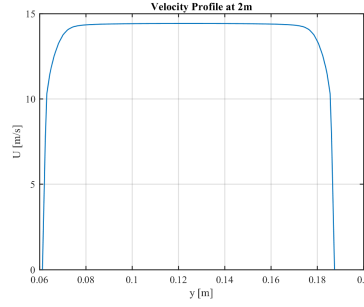


Figure 3.29: x=2m

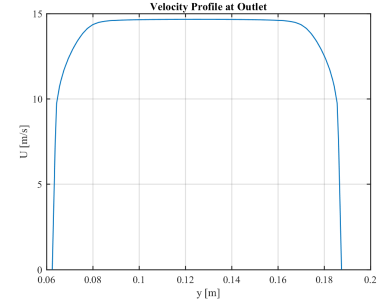


Figure 3.30: Outlet

By integration over the section the mean velocity at the outlet has been computed, $v_{out,m} = 13,8198 \text{ m/s}$. The flow rate and the velocity reduction coefficient are:

$$Q = H_{out} * v_{out,m} = 1,727475 \text{ m}^3/\text{s}; \quad \varphi = \frac{v_{out,m}}{v_{out,is}} = 0,9414$$

The longitudinal distribution of cross-sectional-averaged total,static pressure along the nozzle and velocity are:

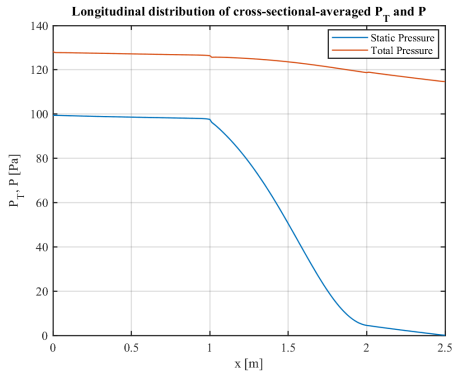


Figure 3.31: $P_{T,m}$ and $P_{s,m}$ distribution

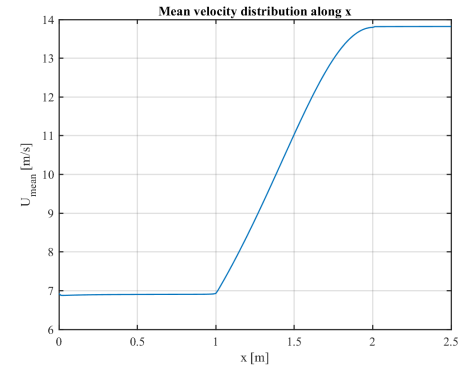


Figure 3.32: Velocity distribution

The distribution of wall shear stress along the Top and Bottom Wall patches is:

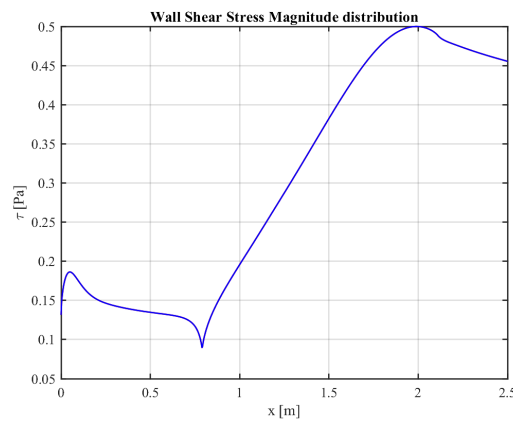


Figure 3.33: τ_w distribution

The stresses increase in the convergent part due to the higher velocity gradient generated by the wall effect.

3.5 Grid Independence

The evaluation of the grid independence of the three simulations has been made by the comparison of the obtained results.

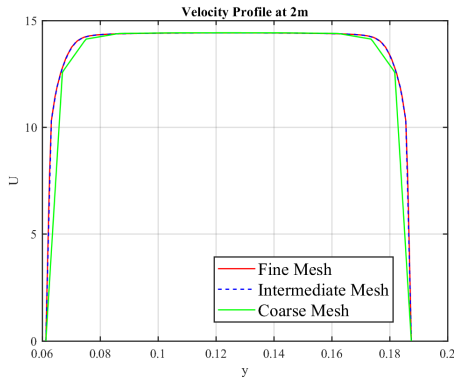


Figure 3.34: velocity at $x=2m$

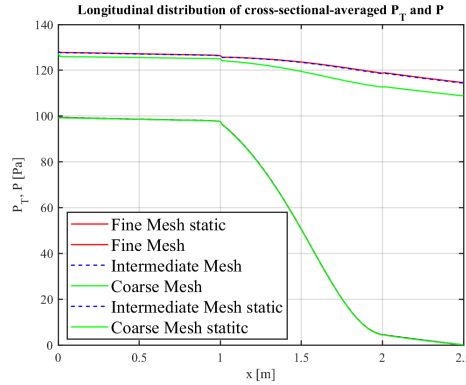


Figure 3.35: Pressure distribution

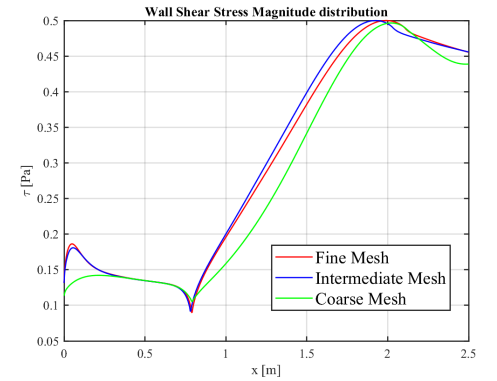


Figure 3.36: Stress distribution

The reported figures clearly show that the Fine and the Intermediate meshes have reached the grid independence; in fact they results are equal and the relative plots almost completely overlap. The coarse mesh results are slightly different and this can be seen from the velocity profile. In this case the difference between the coarse mesh and the other two is more marked than the laminar case, this is probably due to the introduction of the turbulence phenomena. Results are more sensible to the mesh.

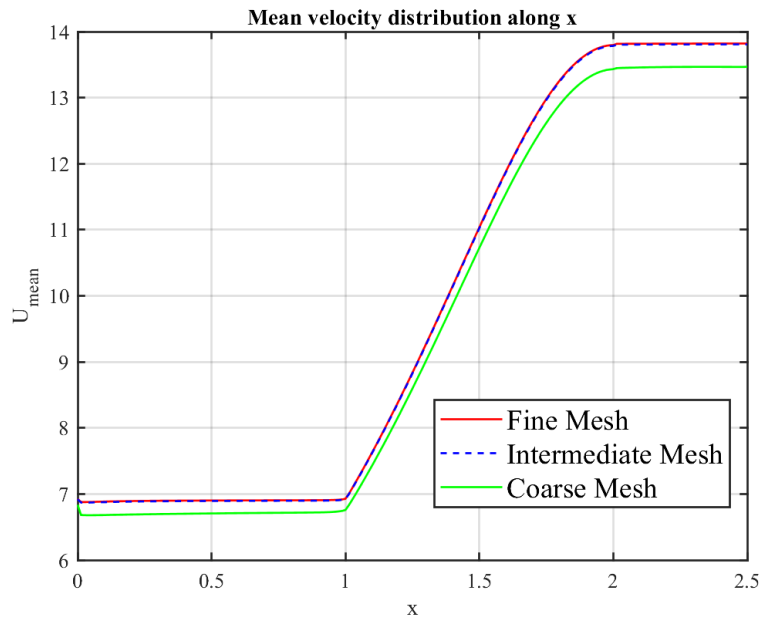


Figure 3.37: Mean U distribution along the nozzle

4 Turbulent Flow: No Wall Function

4.1 Simulation Setting

The third request of the Exercise ask to perform a simulation with a wall resolving Two-equations Turbulent model on the same flow of the previous case, same conditions:

$$k_{in} = 0,202m^2/s^2, \nu_{T,in} = 0,00615m/s^2, P_{T,in} - P_{out} = 129,34MPa, v_{in,is} = 7,34m/s.$$

The model chosen is the $k - \omega SST$ because it can solve up to the wall without singularities, like the $k\Omega$, and it works well in the free flow, like the $k\epsilon$ one. It is specified inside the file **constant/MomentumTransport**:

simulationType RAS, model $k\Omega SST$.

$$\omega = 0.09^{-1/4} \frac{k^{1/2}}{l_T} = 32,8227s^{-1}$$

Boundary conditions for both velocity and pressure fields are of the same type of the laminar case but with the right values of $v_{in,is}$ and $P_{T,in}$. BCs for k, ω, ν_T defined in the **0/** folder are summarized below:

0: BCs	k	ω	ν_T
Top & Bottom Wall	type: kqRWallFunction value: uniform \$inletk	type: omegaWallFunction value: uniform 1e-5	type: nutkWallFunction value: uniform 0.00615
Inlet	type: turbulentIntensityKineticEnergyInlet intensity: 0.05 value: uniform \$inletk	type: turbulentMixingLengthFrequencyInlet mixingLength: 0.025 value: uniform \$inletOmega	type: fixedValue value: uniform 0.00615
Outlet	zeroGradient	zeroGradient	zeroGradient
Front / Back	empty	empty	empty

Table 4.1: 0/k, 0/omega, 0/nut

The BCs chosen for the wall patches are still the ones used for the wall function case but OpenFoam ignores it if the condition on y^+ is satisfied.

SIMPLE solution algorithm is employed with residual control on k and ω too. Convergence is reached when all the residuals fall below 10^{-6} . The new **fvSchemes** and **fvSolution** files are:

system/fvSchemes	Numerical scheme
ddtSchemes	steadyState
gradSchemes:grad(p)	Gauss linear
div(phi,U)	bounded Gauss linearUpwind grad(U)
div(phi,omega)	bounded Gauss upwind
div(phi,k)	bounded Gauss upwind
laplacianSchemes	Gauss linear corrected

Table 4.2: fvSchemes definition

system/fvSolution	solver	smoother	tolerance
p	GAMG	GaussSeidel	10^{-8}
U	smoothSolver	GaussSeidel	10^{-8}
"k—omega"	smoothSolver	GaussSeidel	10^{-8}

Table 4.3: fvSolution definition

4.2 Mesh Generation

In this case, the simulation did not require wall functions; so, the mesh was constructed trying to reach the lowest possible y^+ value. To achieve the target, the mesh was refined in the transversal direction (in particular near the wall). A strong y *SimpleGrading* of 12 has been used near the bottom and the top wall (10 % of space was filled with 40 % of cells on both sides). In order to minimize the computational cost, the number of cells in x-direction was reduced. The mesh has 475 x 60 x 1 cells.

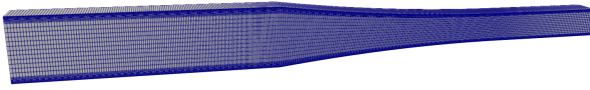


Figure 4.1: mesh, $k-\omega$ SST case

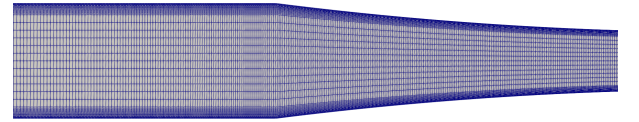


Figure 4.2: mesh zoomed, $k-\omega$ SST case

4.3 Results

The process has reached convergence in 896 iterations. Plots of velocity and pressure are here reported:

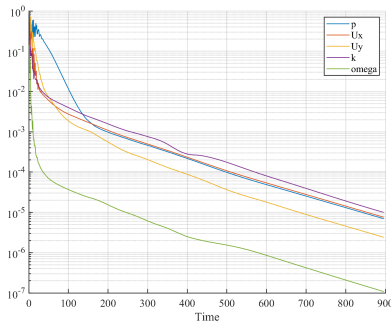


Figure 4.3: Residual trends

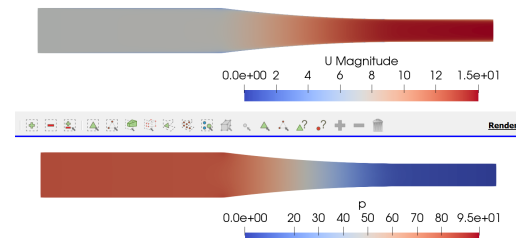


Figure 4.4: U and P

In folder *postProcessing/yPlus/0* at convergence, a maximum value of 0.505 is given, with an average of 0.209 (the values between top and bottom wall are very similar, so only one of them is reported). The upper limit of $y^+ < 1$ is respected. A graph representing y^+ distribution on the patches is here reported:

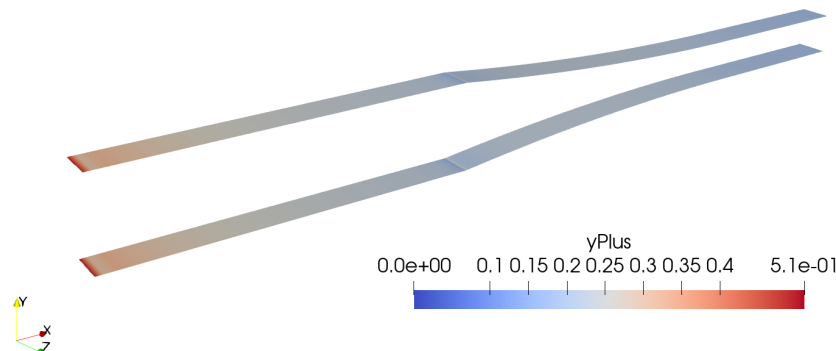


Figure 4.5: y^+ distribution

The velocity profiles in three different sections are reported:

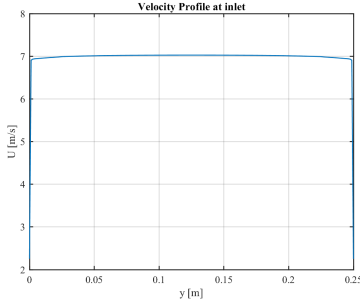


Figure 4.6: Inlet

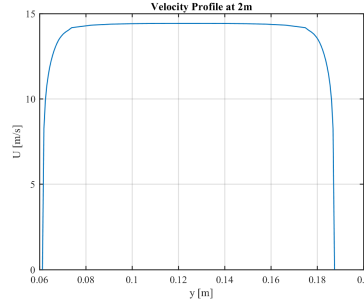


Figure 4.7: x=2m

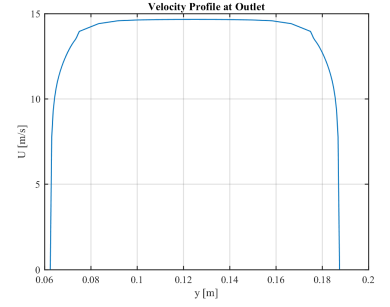


Figure 4.8: Outlet

By integration over the section the mean velocity at the outlet has been computed, $v_{out,m} = 13,8941 m/s$. The flow rate and the velocity reduction coefficient are:

$$Q = H_{out} * v_{out,m} = 1,7367625 m^2/s; \quad \varphi = \frac{v_{out,m}}{v_{out,is}} = 0,9466$$

The longitudinal distribution of cross-sectional-averaged total,static pressure along the nozzle and velocity are:

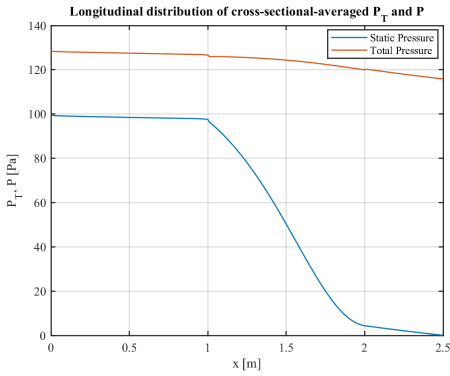
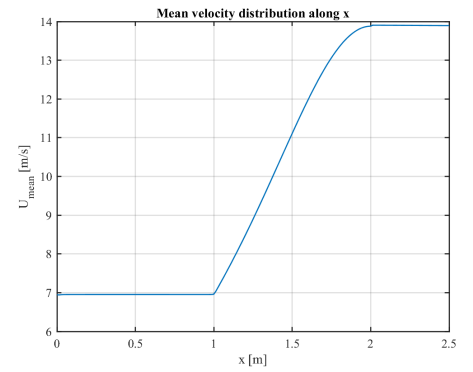
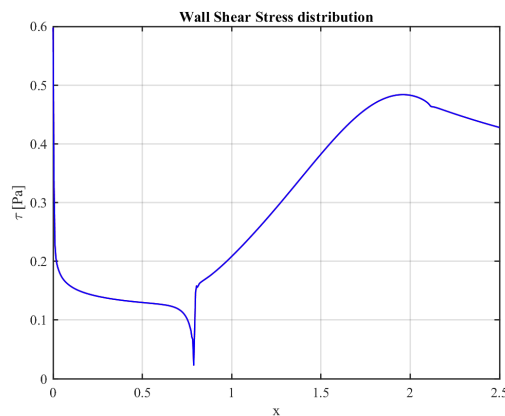
Figure 4.9: $P_{T,m}$ and $P_{s,m}$ distribution

Figure 4.10: Velocity distribution

The distribution of wall shear stress along the Top and Bottom Wall patches is:

Figure 4.11: τ_w distribution

The stresses increase in the convergent part due to the higher velocity gradient generated by the wall effect.

5 Conclusions

Here are presented some considerations about the choices made for the simulations and the results obtained:

- For the discretization of the divergence term $div(phi, U)$ different solutions have been tested such as: *Upwind*, *Linear*, *LimitedLinear*. The best one in terms of convergence speed and quality of the results resulted to be the *linearUpwind Grad(U)*;
- Dealing with a steady state case for the divergences numerical schemes, the *bounded* versions have been used. It increases the Diagonal Dominance of the solution matrix making the equation less difficult to solve;
- Regarding the results, from a physical point of view, the nozzle effect is evident in the graphs of the mean pressures and velocities longitudinal distributions. In the convergent sector the velocity increases at the expense of the pressure that decreases;
- As expected, the velocity profiles clearly show that turbulent flow exhibits its well-known enhanced mixing properties, characterized by a broader and flatter distribution compared to the parabolic profile of laminar flow. This reflects the physical effect of eddy viscosity in the turbulence models.
- The two different turbulent models lead to slightly different results: the relative percentage error between the mean velocities at the exit of the nozzle is about 0,538%. This means that the two models are coherent and they both work well for this case;
- Both in laminar and turbulent NO WF cases the shear stresses distributions present a peak at the beginning, this is probably due to the fact that there the boundary layer is still not completely developed and this information can create problem to the models used in OpenFoam.